



Scratch: Sequencing & Events

Work through each part · Write your answers · Build in Scratch · Circle your tier

30-40 MIN

SCRATCH

AGES 7-13

ALL TIERS

IN-CLASS BUILD

✓ TEACHER ANSWER SHEET — SESSION 3 IN-CLASS ACTIVITY

YOUR NAME

Teacher Copy — Do Not Distribute

TODAY'S DATE

July 1, 2026

GRADE / TIER

All Tiers

PART 1 OF 4 — 10 MIN, NO SCREENS

Warm-Up: Sequence a Morning Routine

🔌 UNPLUGGED PUZZLE — CORRECT SEQUENCE ORDER

Write these numbers in the circles:

- 2 Wake up when the alarm rings (1st)
- 3 Brush your teeth
- 4 Eat breakfast
- 5 Get dressed
- 1 Put on backpack and leave for school (last)

Correct order: 2 → 3 → 4 → 5 → 1

Now answer these:

1. What event starts your routine?

✓ The alarm ringing — a trigger event, just like when green flag clicked in Scratch.

2. What happens if you swap steps 2 and 4?

✓ You eat breakfast before brushing teeth — wrong result. Order matters in algorithms; wrong order = bug.

3. In Scratch, “alarm ringing” is which block?

✓ An Events (yellow) hat block — e.g. when green flag clicked.

PART 2 OF 4 — 10 MIN

Vocabulary & Concept Check

✂ MATCH — ANSWER KEY

A. Sequence

B. Event

C. Sprite

D. Script

E. Stage

A character or object you program in Scratch

C Sprite

Steps carried out one after another, in order

A Sequence

A trigger that tells your program to start running

B Event

Background canvas where sprites live (480×360 px)

E Stage

A stack of blocks that runs when an event fires

D Script

All 5 correct = full 2 pts. Key: C, A, B, E, D

✓ TRUE OR FALSE — ANSWER KEY

Scratch runs all blocks **at the same time** when flag clicked.

F

FALSE — Scratch runs top-to-bottom in sequence (per script). Not simultaneous.

A sprite can have more than one script attached to it.

T

TRUE — each script runs when its own hat block fires independently.

A block NOT connected to a hat block will still run automatically.

F

FALSE — disconnected blocks never run; they need a hat block above them.

Events are yellow hat blocks in Scratch's palette.

T

TRUE — all event blocks are yellow/orange hat-shaped blocks.

Changing the order of blocks always gives the same result.

F

FALSE — order is critical; swapping blocks changes (usually breaks) the output.

3 pts. Answer key: F, T, F, T, F

📦 BLOCK SORT — ANSWER KEY

when clicked

when [space] key pressed

move 10 steps

go to x:0 y:0

say [Hello!] for 2 secs

set size to 100%

✂ Events (Yellow)

● Motion (Blue)

● Looks (Purple)

✓ when ⚑ clicked	✓ move 10 steps	✓ say [Hello!] for 2 secs
✓ when [space] key pressed	✓ go to x:0 y:0	✓ set size to 100%

PART 3 OF 4 — 10 MIN

Trace the Script & Spot the Bug

TRACE THE SCRIPT

```
when ⚑ clicked
move 50 steps
say "Hello!" for 2 secs
move 50 steps
say "I moved!" for 2 secs
```

What does the sprite do?

Step 1: Sprite moves right 50 steps (50px right).

Step 2: Speech bubble says "Hello!" for 2 secs.

Step 3: Sprite moves right another 50 steps (100px total).

Step 4: Says "I moved!" for 2 secs. Script ends.

Key: Sequential, top-to-bottom. No steps happen at once.
Total movement = 100px right.

SPOT THE BUG — ANSWER KEY

```
move 10 steps ← no hat block! ⚑
say "Yay!" for 2 secs
when [up arrow] key pressed
change y by 20
```

What is the bug?

✓ **move 10 steps** and **say "Yay!"** are NOT connected to a hat block — they float and never run.

Corrected script (correct order):

```
when [up arrow] key pressed ← HAT BLOCK
FIRST ✓
change y by 20
say "Yay!" for 2 secs
```

Rule: Every script MUST start with a hat block. No hat block = script never runs. Most common Scratch beginner bug!

PART 4 OF 4 — 30-40 MIN

Open Build — Choose Your Tier & Build!

☆ TIER A • GR 1-3

Starter cat sprite, green flag, move + say.

```
when ⚑ clicked
move 10 steps
say "Hello!" for 2 secs
```

✓ **Goal:** sprite moves AND says something when flag clicked.
Buddy pairs swap & test.

🛠 TIER B • GR 3-5

Own sprite, 2 scripts with different events.

✓ **Script A (when ⚑ clicked):** 3 actions.
Script B (when [space] pressed): 1 action.
Plan both events + actions BEFORE touching keyboard.
Project named & saved.

2 separate scripts, each triggered independently.

🔪 TIER C • GR 6-9

2 sprites, each with its own event and sequence.

✓ **Sprite 1:** when ⚑ clicked → 3 actions.
Sprite 2: when [right arrow] → 3 actions.
Challenge: pressing right arrow fires Sprite 2 only. Both CAN fire simultaneously if events overlap. This is parallel execution.

🎫 EXIT TICKET — ANSWER KEY

1. In your own words: what is a sequence?

✓ "A list of steps that happen one at a time, in exact order. Change the order and the result changes — or breaks! In Scratch, blocks in a script run in a sequence: top to bottom."

2. Difference between a script and an event?

✓ **A script = the full stack of blocks (all instructions for a sprite). An event = the hat block at the TOP that starts it running. No event = script never runs.**

3. Point to one block and say "This block is here because..."

✓ **Open-ended. Award full credit if student points to any block and gives a purpose explanation that shows understanding of sequencing or events.**

ACTIVITY RUBRIC

Section	What to Look For	Pts
Part 1 Warm-Up	Correct order 2→3→4→5→1; event & swap questions answered with reasoning	3
Part 2 Vocab Match	All 5 letters correct: C, A, B, E, D	2
Part 2 True/False	All 5 T/F correct: F, T, F, T, F. Accept brief reasoning for partial credit.	3
Part 2 Block Sort	6 blocks in correct columns. Events: 2 yellow. Motion: 2 blue. Looks: 2 purple.	2

Part 3 Trace & Bug	4 steps described correctly; bug identified (no hat block); corrected script written.	3
Part 4 Build	Tier goal met; project runs; exit ticket answered; project named. Functionality > aesthetics.	7
Total		20

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